MILITARY SPECIFICATION SHEET

CABLE ASSEMBLY, RADIO FREQUENCY (AIRCRAFT AUDIO), CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of specification MIL-C-22442.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for audio frequency cord assemblies with braided and molded elastomer jackets.

1.2 Military part number.

Molded Elastomer Jacket Coil Cord M22442/35-1

Molded Elastomer Jacket Straight M22442/35-2

Braided Jacket M22442/35-3

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-55668	Cord, Electrical; Audio Subminiature (Retractile and Straight)

STANDARDS

MILITARY

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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STANDARDS (Continued)

MILITARY (Continued)

MIL-STD-454	Standard General Requirements for Electronic Equipment

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL) STANDARDS

UL-94	Standard for Flame Retardance

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050 or 333 Pfingsten Road, Northbrook, IL 60062.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Fungus-resistant materials. Cords shall be made of fungus-resistant materials as specified in Requirement 4 of MIL-STD-454.

3.2 Design and construction.

3.2.1 Conductors. The nominal area of the conductors shall be 320 circular mils and the minimum 310 circular mils. Each conductor shall consist of 37 bunch-stranded AWG #40 (0.003145" dia) tinned cadmium-bronze wires. The strands shall be twisted to have a left hand lay of 1 inch nominal. The individual strands, before bunching and tinning, shall have a minimum tensile strength of 90,000 pound-force per square inch and have a conductivity grade of at least 80 percent.

3.2.1.1 Tensile strength. The tensile strength of the wire shall not decrease more than 10 percent after tinning and insulating when tested in accordance with 4.1.

3.2.2 Insulation. The insulation shall be melt-extruded of Teflon FEP having a minimum wall thickness of .006 inches.

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3.2.3 Color code. For circuit identification, the insulation for the M22442/35-1 and M22442/35-2 cord assembly shall be colored white, black, red, green, black, and white in that order arranged in accordance with Figure 1. The insulation for the M22442/35-3 cord assembly shall be colored red, green, white, orange and black arranged in accordance with Figure 2.

3.2.4 Stay cords. The stay cords shall be composed of nylon with a breaking strength of 20 pounds when braided together; they shall not abrade or otherwise damage the insulation of the conductors when the cord is bent or stretched.

3.2.5 Jacketing materials.

3.2.5.1 Braided. The cord shall be covered with a braided jacket composed of black nylon, 210 denier, 3 ends, 20 carrier, approximately 30 picks per inch.

3.2.5.2 Molded elastomer. The jacket shall be a smooth, black elastomer formed free of surface irregularities in accordance with MIL-C-55668. The jacket O.D. shall be in the range of .168 to .180 diameter with a jacket wall of .016 to .025. The jacket shall be able to withstand the vertical burn test of Underwriters' Laboratory specification UL-94 class rate V-1 non-dripping.

3.2.6 Shield. The shielding will be wrapped shielding, 85-95 percent coverage, nickel plated soft drawn copper.

3.3 Performance.

3.3.1 Temperature. The cord shall be flexible and resilient throughout the temperature range of -55°C to +85° and shall not show evidence of cracking or other damage as a result of exposure to the temperature range specified.

3.3.2 Flex life. The cord assembly shall have a mean flex life of 50,000 flexes without showing evidence of:

a. Damage to the outside jacket.

b. Damage to the insulation of the individual conductors.

c. Electrical discontinuity.

3.3.3 Electrical characteristics.

3.3.3.1 Dielectric withstanding voltage. There shall be no evidence of breakdown when the cords are tested as specified in 4.5. Leakage currents shall be recorded and shall not vary more than 30 percent from initial values.

3.3.3.2 Insulation resistance. When tested as specified in 4.6, the insulation resistance shall not be less than 100 megohms.

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3.3.3.3 Electrostatic and electromagnetic isolation. The cord shall meet the electrostatic and electromagnetic isolation requirements listed in Table I and when tested in accordance with Figure 3 and 4.7.

TABLE I. Isolation requirements.

Frequency Hz	Electrostatic isolation (minimum decibels)	Electromagnetic isolation (minimum decibels)
20,000	106	75
10,000	112	81
5,000	118	87

3.3.4 Continuity. Each conductor in the cord shall be continuous when tested as specified in 4.8 and shall not exceed 4 millivolts drop above that of an equivalent length of the same conductor with a test current of 1.6 amperes (dc).

4. QUALITY ASSURANCE PROVISIONS

4.1 Tensile strength. The wire shall be tested to meet the requirements of 3.2.1.1. One strand selected at random from a conductor shall be tested.

4.2 Molded elastomer. The molded elastomer shall be tested in accordance with the vertical burn test of Underwriters Laboratory specification UL-94 class rate V-1 non-dripping.

4.3 Temperature. The unterminated cord shall be placed within a test chamber wherein a temperature of 85°C (185°F) is maintained for a period of four hours. The cord shall be wrapped about a mandrel while at this temperature without damage to the cord jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cord diameter. The cord shall be attached to the mandrel and suspended vertically with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping without handling. The cord shall be inspected for damage or deformation and then subjected to a test chamber wherein a temperature of minus 55°C (minus 67°F) is maintained. After 20 hours of storage at -55°C, the cord shall be wrapped about the mandrel, while at this temperature, in the same manner as it was at the positive extreme temperature. The cord shall be examined for cracks or any other damage to the jacket or conductor insulation. The cord shall meet the requirements of 3.3.1.

4.4 Flex life. Five samples of the unterminated cord shall be clamped and suspended through holes in a metal bar having a rectangular cross section of 1/2 inch by 5/8 inch as described below:

a. The clamping point shall be immediately above the metal bar; the cord clamp shall rest on top of the metal bar.

b. The cord shall be clamped in such a manner that it will not turn in the hole during the flexing test.

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c. The length of the through holes in the metal bar shall be 5/8 inch. The diameters of the through holes shall be 0.005 to 0.010 in. greater than the diameter of the cord samples.

d. The through holes in the metal bar shall incorporate a 1/8 inch radius at the end from which the cord emerges (bending point).

e. A three-pound weight shall be hung from the cord at a point 8 to 12 inches below the bar to assure that the cord is held in a vertical position.

f. A small low-current drain, a 6-volt lamp, and a suitable power source shall be connected in series to each conductor to indicate electrical continuity (see Figure 4). The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetical mean of the flexing cycles of the five samples of the cord before electrical discontinuity occurs shall be taken as the flexing life of the cord (see 3.3.2).

4.5 Dielectric withstanding voltage. Dielectric strength shall be performed in accordance with Method 301 of MIL-STD-202. A potential of 600 volts, 60 Hz, shall be applied for one minute. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. The cord shall meet the requirements of 3.3.3.1.

4.6 Insulation resistance. Insulation resistance shall be in accordance with Method 302 of MIL-STD-202, test condition B. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. The cord shall meet the requirements of 3.3.3.2.

4.7 Electrostatic and electromagnetic isolation. The cord shall be tested for electromagnetic and electrostatic isolation to determine compliance with 3.3.3.3 and Table I when tested in accordance with Figure 3.

4.8 Continuity. One and six tenths (1.6) amperes of direct current shall be passed through each conductor of each cord. The voltage drop across the conductor shall not exceed that specified in 3.3.4.

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TABLE II. Cable inspections.

Inspection	Paragraph		Group I	Group II
	Require-ments	Test	Qualifi-cation tests	Sample testing per 4.5.1
Tensile Strength	3.2.2	4.1	X	X
Molded Elastomer	3.2.5.2	4.2	X	X
Temperature	3.3.1	4.3	X	X
Flex Life	3.3.2	4.4	X	X
Dielectric Withstanding Voltage	3.3.3.1	4.5	X	X
Insulation Resistance	3.3.3.2	4.6	X	X
Electrostatic and Electromagnetic isolation	3.3.3.3	4.7	X	X
Continuity	3.3.3.4	4.8	X	X

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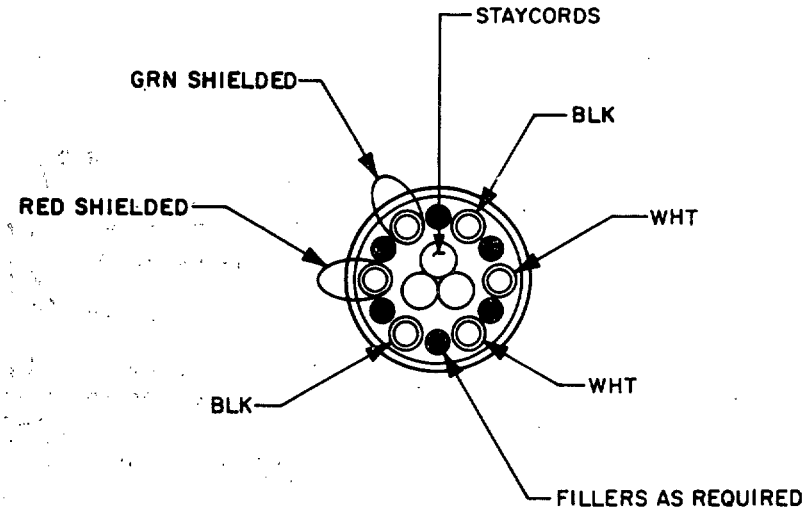


FIGURE 1. Cable construction.
P/N M22442/35-1 and P/N M22442/35-2

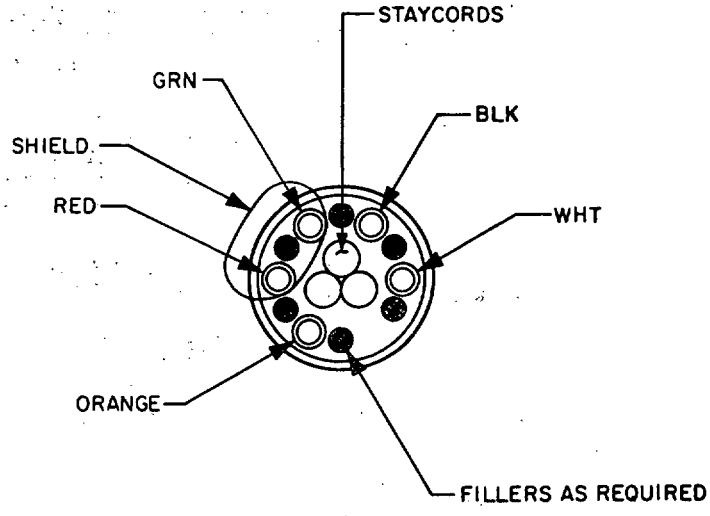
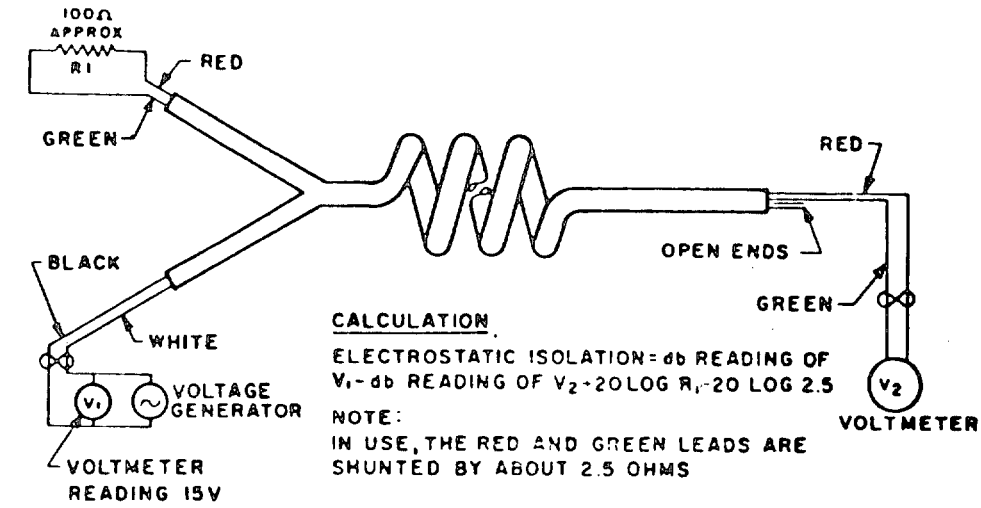
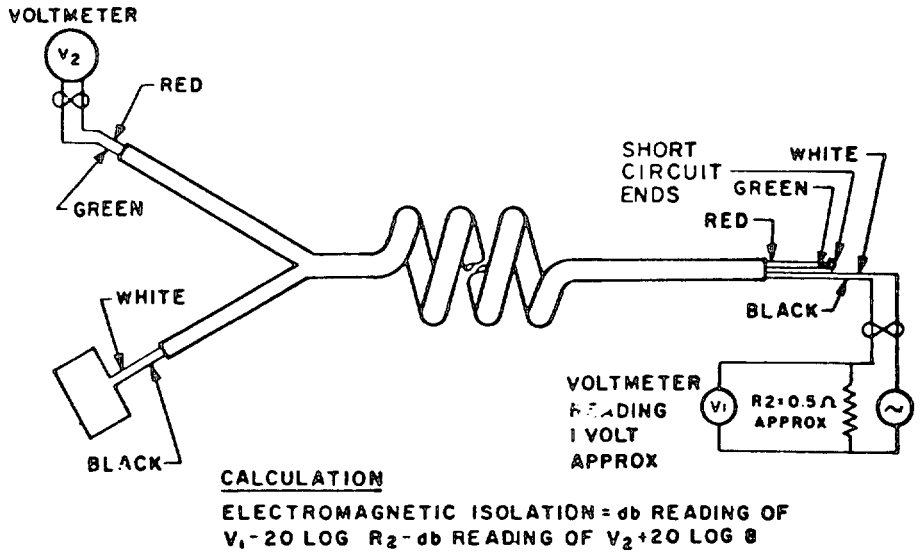


FIGURE 2. Cable construction.
P/N M22442/35-3

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ELECTROSTATIC ISOLATION TEST



ELECTROMAGNETIC ISOLATION TEST

NOTE: Test setup applies to straight cords also.

FIGURE 3. Test set-up.

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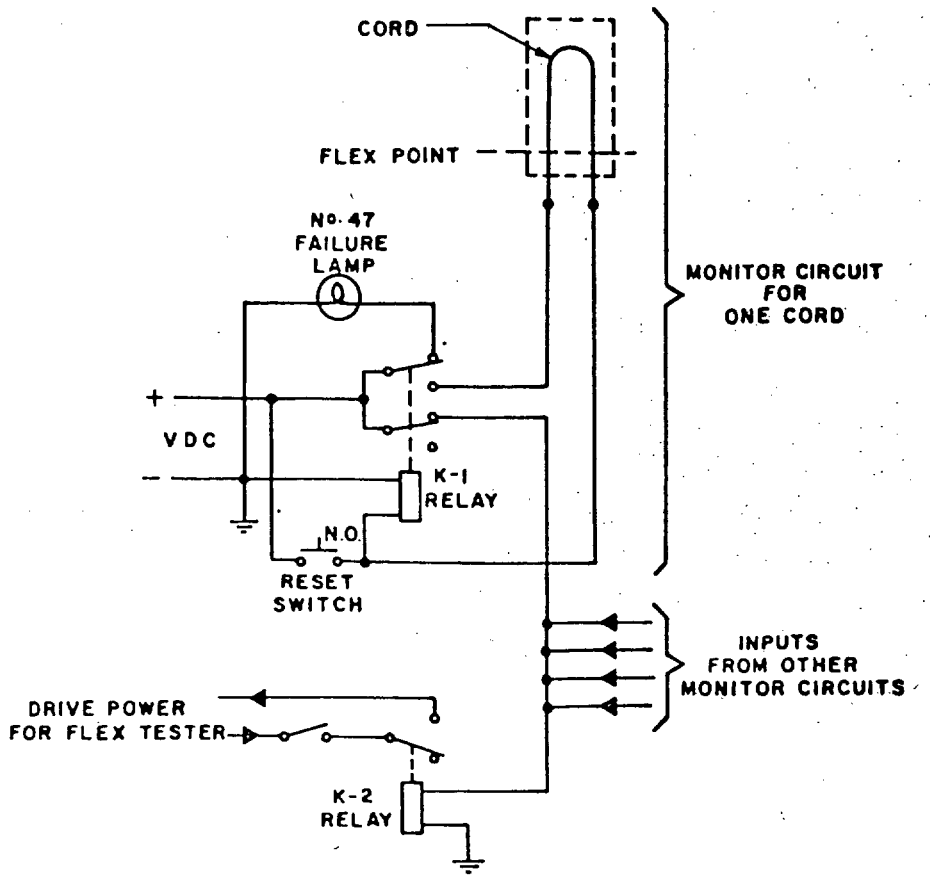


FIGURE 4. Monitor circuit for cord flex tester.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22442/35	CABLE ASSEMBLY, RADIO FREQUENCY (AIRCRAFT AUDIO), CABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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